GRADE 7 · COMPUTER SCIENCE · SECOND TERM

The Computer Study Guide 💻

Every question on your worksheet with its answer, the 25 written answers to learn, memory tricks and a cheat sheet. With Chip the Mouse 🖱️.



PART 1 · LESSONS 1–4
Web pages with HTML

PART 2 · LESSONS 5–6
Styling with CSS

PART 3 · LESSONS 7–8
Programming and Python

PART 4 · LESSONS 9–10
Drawing in GIMP

🧭 How to use this guide

There is one section for each of the 10 Computer lessons. Each one has **the big idea**, the **key words**, a picture, **every worksheet question** of that lesson with its answer, the **written answers** to learn, a **memory trick**, a **Watch out** box and **Try these** questions (answers at the back). At the end: a one-page **cheat sheet** and **all 25 model answers** together.

💡 **MEMORY TRICK**
An easy way to remember.

⚠️ **WATCH OUT**
A mix-up that loses marks.

✏️ **TRY THESE**
Practise, then check.

📝 Exam tips

- **Learn the 25 written answers** (Q1–Q25) word for word: they are the biggest part of the exam. Use the Answer Trainer: ten a day.
- **Look for the keyword** in each question (in yellow in this guide). It tells you which answer to write: "HTML tags" is not the same answer as "HTML", and "script mode" is not "Interactive Mode".
- Write the **key phrases** (in green): they are what the teacher looks for. Numbered answers (Q3, Q9, Q12, Q21) need every item.
- **Tick the correct option:** read all three options first. "Not" questions (A3, A6, A8, A10) want the odd one out.
- **True or false:** one wrong word makes a sentence false: "CSS *creates* web pages" (F), "Gradient *darkens*" (F), "Filters *select*" (F).

PART 1 · WEB PAGES WITH HTML

Lesson 1 · What is HTML?

THE BIG IDEA

HTML (Hypertext Markup Language) is the language web pages are written in. It was developed in **1990** by the scientist **Tim Berners-Lee**. An HTML file is just plain text, so you can type it in **Notepad** and open it in any **web browser**.

KEY WORDS: **HTML** · **web page** · **World Wide Web** · **hyperlink** · **W3C** · **HTML editor** · **text editor** · **web browser**

index.html - Notepad

```
<HTML>
 <BODY>
  Hello, world!
 </BODY>
</HTML>
```

1. Type the HTML in a text editor and save it as **.html**

Microsoft Edge

Hello, world!

2. Double-click the file, or open Edge and press **Ctrl + O**

From your worksheet

#	Fill in the blanks
B1	**W3C** is an organisation which sets international standards for the World Wide Web.
B3	**HTML** documents can be viewed on any type of web browser.

#	Write T for true and F for false	Ans
C2	You can create HTML documents with Notepad, WordPad, or any other text editor. HTML is just text, so any text editor works.	**T**

Written questions (learn these answers)

Q1 What do you know about HTML?

> HTML stands for **Hypertext Markup Language**. It was developed in the year **1990** by the scientist **Tim Berners-Lee**.

💡 **H**ey **T**im, **M**ake **L**inks! Tim Berners-Lee, 1990.

Q2 What's the use of HTML?

> HTML is mainly used to **create web pages** that are displayed on the **World Wide Web**. Each page contains a series of **connections to other pages** called **hyperlinks**.

💡 Think of the spider: HTML makes the **pages** 📄, **hyperlinks** join them into the **web** 🕸️.

Q3 Write any three features of HTML 5?

> 1. HTML documents are **simple text files**.
> 2. HTML is **not a case sensitive** language.
> 3. HTML has **many tags** to make web pages.

💡 Three features: **Text**, **Case**, **Tags**. Picture a **text** file in a suit**case** covered in **tags**! 🧳

Q4 **Define HTML Editors and Text Editors.**

> We can create an HTML document using an **HTML editor** or any **text editor**.
> HTML editors are **Adobe Dreamweaver**, **UltraEdit**, **TextMate**.
> Text editors are **Notepad**, **Notepad++**.
> We can create our HTML documents using Notepad.

💡 Fancy editors have fancy names (**Dream**weaver, **Ultra**Edit, Text**Mate**). Plain text editors have plain names: **Notepad**! 🗒️

Q5 **How can we view the HTML document?**

> After creating and saving an HTML document we can view it by **double clicking** on the HTML document.
> We can also open an HTML file by opening **Microsoft Edge** by clicking on its icon on the desktop or the taskbar and then pressing **Control + O** keys.

💡 Two ways to view: **double-click** the file 🖱️🖱️, or open **Edge** and press **Ctrl + O** (O for **Open**).

⚠️ **WATCH OUT**

Notepad is a *text* editor; **Dreamweaver**, **UltraEdit** and **TextMate** are *HTML* editors. Don't swap them. **W3C** (World Wide Web Consortium) sets the rules for the web; it is not a program.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Who developed HTML, and in which year?
2. Name two text editors you can use to write HTML.
3. Which two keys open an HTML file in Microsoft Edge?



PART 1 · WEB PAGES WITH HTML

Lesson 2 · HTML tags and page structure

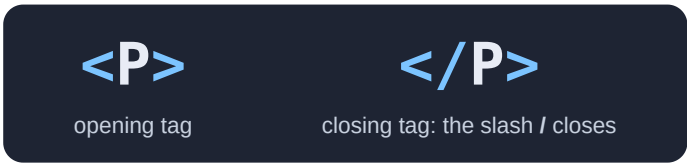
THE BIG IDEA

Tags are the instructions of HTML. Each tag sits inside **angular brackets** < >. Most tags come in pairs: an **opening tag** like `<BODY>` and a **closing tag** with a slash like `</BODY>`; these are **container tags**. Every page has the same skeleton: HTML on the outside, then HEAD, then BODY.

KEY WORDS `tag` `angular brackets < >` `opening tag` `closing tag` `container tag` `HEAD` `BODY`



The page skeleton: boxes inside a box



< is "less than", > is "greater than": together, angular brackets

From your worksheet

#	Tick the correct option
A1	**Which of the following is the correct structure of HTML tags?** (a) `<HTML> <HTML> </HEAD> <HEAD> </BODY> </BODY>` (b) `<HTML> </HTML> <HEAD> </HEAD> <BODY> </BODY>` ✔ (c) `<HTML> <HEAD> </HEAD> <BODY> </BODY> </HTML>`

Written questions (learn these answers)

Q6 **What do you know about HTML tags?**

> Tags are **elements** that are used to specify the **structure** of an HTML document. HTML tags begin with the **less than** character (<) and end with the **greater than** symbol (>), which are called **angular brackets**.

💡 A tag is a word in a pointy sandwich 🥪: **<** (**L**ess than) at the start, **>** (**G**reater than) at the end. Both together are **angular brackets**: they have angles! 📐

Q7 **What is the difference between opening and closing tags?**

> In HTML, most tags have an **opening** and a **closing tag**. For example **<HTML>** is the opening tag and **</HTML>** is the closing tag. The tags which have both opening and ending tags are called **container tags**.

💡 The closing tag has a */*: a door shutting 🚪. A box with a top and a bottom holds things inside: a **container** 📦. Both tags = **container tags**.

💡 **MEMORY TRICK**

H-H-B-H: open **H**TML, then **H**EAD (open and close), then **B**ODY (open and close), and close **H**TML last. The head comes before the body, like on you! 🙂

⚠️ **WATCH OUT**

In A1, option (b) closes HTML straight away, so HEAD and BODY end up outside the page. `</HTML>` is always the **very last** tag. The slash goes at the **start**: `</BODY>`, not `<BODY/>`.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Write the closing tag for `<HEAD>`.
2. Which tag is always the last tag on the page?
3. What do we call tags that have both an opening and a closing tag?



PART 1 · WEB PAGES WITH HTML

Lesson 3 · Pictures on a web page

THE BIG IDEA

The `<IMG>` tag adds a picture to a web page. Its **attributes** give extra details: **src** (which picture file), **alt** (the alternate text shown if the picture can't be displayed), **height** and **width**. Pictures can be **.jpg**, **.gif**, **.png** or **.bmp** files.

KEY WORDS: **<IMG>** · **attribute** · **src** · **alt (alternate text)** · **height** · **width** · **.jpg .gif .png**

```
<IMG src="cat.png" alt="My cat" height="200" width="300">
```

▲ the tag ▲ the picture file ▲ text if the picture can't show ▲ size

Picture file? ✔	Not a picture ✘
.jpg .gif .png .bmp	.xlsx (Excel) .docx (Word)

🖼️❌ *My cat*

If the picture can't load, the **alt** text is shown

From your worksheet

#	Tick the correct option
A6	**Which of the following is not an image file format?** (a) .jpg (b) .gif **✔ (c) .xlsx**
A8	**Which of the following is not an attribute of the `<IMG>` tag?** (a) Height (b) Width **✔ (c) Target**

#	Fill in the blanks
B6	The **<IMG>** tag is used to add a picture to a web page.
B7	The **alt** attribute is used to specify an alternate text.

#	Write T for true and F for false	Ans
C6	You cannot insert .png images on a web page. .png is a picture format, so you **can** insert it.	F
C9	Alt attribute is used to specify the text if an image is not displayed on a web page.	T

💡 **MEMORY TRICK**

alt = **alt**ernate text: the picture's **alt**ernative when it can't be seen. **.xlsx** has an **x** for e**x**cel: a spreadsheet, not a picture.

⚠️ **WATCH OUT**

target is an attribute of **links**, not of `<IMG>`. The sentence "You cannot insert .png images" is **False**: .png **can** be inserted.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which attribute gives the text shown when a picture cannot be displayed?
2. Is .gif a picture format? Is .xlsx?
3. Write an `<IMG>` tag for a picture called **sun.jpg**.

PART 1 · WEB PAGES WITH HTML

Lesson 4 · Links: internal, external and URLs

THE BIG IDEA

A **link** (hyperlink) takes you to another page when you click it. There are two types of linking: **internal** and **external**. The address of a page is its **URL**: an **absolute URL** is the complete address; a **relative URL** is a short one. The **target** attribute says where the link opens, and **link**, **alink** and **vlink** set link colours.

KEY WORDS: **link (hyperlink)** · **internal linking** · **external linking** · **URL** · **absolute URL** · **relative URL** · **target** · **link · alink · vlink**

`https://` `www.myschool.com` `/pages/` `news.html`

▲ how ▲ the website ▲ the folder ▲ the file

An **absolute URL** has every part: the complete path. A **relative URL** is just the end, e.g. **news.html**.

Attribute (in <BODY>)	Colours…
link	links not clicked yet
alink	**a**ctive links (being clicked)
vlink	**v**isited links

```
<A href="news.html"
target="_blank">News</A>
```

target="_blank" opens the link in a new tab

From your worksheet

#	Tick the correct option
A7	**Which of the following is a type of linking?** (a) Internal linking (b) External linking ✔ **(c) Both (a) and (b)**

A9	**Which of the following is used to change the colour of a link?** (a) Alink (b) Vlink **✔ (c) All of these**

#	Fill in the blanks
B8	The **target** attribute is used to specify a target.
B9	**Absolute URL** includes a complete description of the file.

#	Write T for true and F for false	Ans
C7	Internal linking is used to link a page to another page. In the exam write **T**. Many books say internal links go to another page of the same website, or another part of the same page.	**T**
C8	An absolute URL refers to the complete path of a file.	**T**

💡 **MEMORY TRICK**

Absolute = **A**ll of the address. **a**link = **a**ctive, **v**link = **v**isited. When a question says "All of these" or "Both", check that every option is true!

⚠️ **WATCH OUT**

target belongs to a **link**, not to a picture. Internal = links inside your own website; external = links to a **different** website.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which attribute sets the colour of visited links?
2. Is **news.html** an absolute URL or a relative URL?
3. Name the two types of linking.

PART 2 · STYLING WITH CSS

Lesson 5 · What is CSS?

THE BIG IDEA

CSS stands for **Cascading Style Sheets**. HTML builds the page; CSS **styles** it: colours, fonts, sizes and positions. CSS does **not** create pages. A CSS file is saved with a **.css** extension, and one file can style a whole website, so large websites are easy to format.

KEY WORDS: CSS | Cascading Style Sheets | style | selector | property | value | .css

```
h1 { color: blue; }
```

selector	property	value
which elements	what to change	change it to what

A CSS rule has three parts: **selector**, **property** and **value**. "Make every h1 heading blue."

HTML builds the page + **CSS** styles the page

style.css → every page of the website

Change one file, restyle the whole website

From your worksheet

#	Tick the correct option
A2	**Which of the following is a full form of CSS?** ✔ **(a) Cascading Style Sheets** (b) Common Style Sheet (c) Common Selector Sheet

#	Fill in the blanks
B4	A CSS rule is made up of three parts, a selector, property and **value**.
B5	CSS files are saved with a **.CSS** extension.

#	Write T for true and F for false	Ans
C1	CSS is used to create web pages. HTML creates web pages; CSS **styles** them.	F
C3	It is easy to format large websites using CSS.	T
C4	CSS selectors are used to manipulate the HTML elements in a web page. The selector picks (selects) which HTML elements get the style.	T

Written questions (learn these answers)

Q8 What do you know about CSS?

> CSS stands for **Cascading Style Sheets**. CSS is used to **style HTML pages** and it's used to specify **how HTML elements are displayed**.

💡 HTML builds the house 🏠, CSS paints and decorates it 🎨. A **cascade** is a waterfall: the styles flow down onto every page.

💡 **MEMORY TRICK**

S-P-V: "**S**elect, **P**ick, **V**alue": the **S**elector chooses who, the **P**roperty says what, the **V**alue says how.

⚠️ **WATCH OUT**

CSS **styles** pages but does **not create** them (C1 is False). The full form is **Cascading** Style **Sheets**, with an s.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. In `p { color: red; }` name the selector, the property and the value.
2. What extension does a CSS file have?
3. True or false: CSS is used to create web pages.



PART 2 · STYLING WITH CSS

Lesson 6 · Three kinds of style sheet

THE BIG IDEA

There are three types of CSS: **inline** (a **style attribute** inside one tag, for a single element), **internal** (a `<STYLE>` block in the **head section**, for one page) and **external** (a separate **.css** file, for many pages). Properties like **font-family**, **text-indent** and **background-position** do the styling.

KEY WORDS: **inline** · **internal** · **external** · **style attribute** · **head section** · **font-family** · **text-indent** · **background-position**

1 · Inline

```
<P
style="color:red">Hi</P>
```

In **one tag**: styles a **single element** with the **style attribute**.

2 · Internal

```
<HEAD>
 <STYLE>
  p {color:red}
 </STYLE>
</HEAD>
```

In the **head section**: styles **one page**.

3 · External

```
📄 style.css
p {color:red}
```

In a **separate .css file**: styles **many pages**.

Property	What it does	Example
font-family	sets the font of the text	`font-family: Arial;`
text-indent	indents the first line of text	`text-indent: 30px;`
background-image	puts a picture behind the page	`background-image: url("sky.jpg");`
background-position	sets the position of that image	`background-position: center;`

From your worksheet

#	Tick the correct option
A3	**Which of the following is not a type of style sheet?** (a) External (b) Internal **✔ (c) Inside**
A4	**___ is used to set the font of a text.** **✔ (a) Font family** (b) Font name (c) Font style
A5	**Which of the following is used to specify the position of an image on the web page?** (a) Background-image (b) Background **✔ (c) Background-position**

#	Fill in the blanks
B2	The **text-indent** property is used to set the indentation of the first line of text.

#	Write T for true and F for false	Ans
C5	The internal style sheet style is defined in the head section of a web page.	T

Written questions (learn these answers)

Q9 **List down the different types of CSS?**

1. **Inline** style sheet
2. **Internal** style sheet
3. **External** style sheet

💡 Small, medium, large: **In**line (one line), **In**ternal (one page, in the head), **Ex**ternal (a whole website, in a separate .css file). In, In, Ex! 📏

Q10 **Define inline style sheet.**

The inline style sheet is used to apply the style to a **single HTML element** using the **style attribute**.

💡 **In**line = **in** the **line** of ONE element ☝️: the **style attribute** sits right inside that one tag.

⚠️ **WATCH OUT**

"**Inside**" sounds right but is **not** a type: the three are **In**line, **In**ternal, **Ex**ternal. The font is set by **font-family** (not "font name"). **background-image** chooses the picture; **background-position** places it.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which type of style sheet uses the **style attribute**?
2. Which property sets the indentation of the first line of text?
3. Where is an internal style sheet written?

Lesson 7 · Programs and programming languages

THE BIG IDEA

A **program** is a sequence of instructions that tells the computer how to do a particular task. Programs are written in **programming languages**, and writing them is called **programming**. There are three main types of language: **machine**, **assembly** and **high level**. A computer only understands machine language, so a **translator** (a **compiler** or an **interpreter**) converts high level code into machine code.

KEY WORDS: **program** | **programming** | **programming language** | **machine language** | **assembly language** | **high level language** | **translator** | **compiler** | **interpreter**

3 · High level `print("Hi")`
almost English: easy for people

2 · Assembly level `MOV A, 5`
short code words

1 · Machine level `10110000 0101`
only 0s and 1s: what the computer understands

The three types of programming language

`print("Hi")` → 🗣️ **translator** → `0101 1100`
high level code — **compiler or interpreter** — machine code

Compiler	Interpreter
translates the **whole** program in one go	translates and runs it **one line at a time** (Python uses one)

This lesson has no tick, blank or true/false questions on the worksheet: just the three written answers below.

Written questions (learn these answers)

Q11 What do you know about computer programming?

A **sequence of instructions** given to the computer to perform a **particular task** is called a **program**. These programs are created using **programming languages**. The process of creating a program is also known as **programming**.

💡 A program is a recipe 📋: a **sequence of instructions** for one **task**. The language you write it in is a **programming language**; writing it is **programming**.

Q12 Write the names of types of computer programming languages?

There are three main types:

1. **Machine** level language
2. **Assembly** level language
3. **High** level language

💡 Climb the ladder 🪜: **M**achine (0s and 1s), **A**ssembly (short codes), **H**igh (almost English). **M-A-H**: "Ma, help!"

Q13 Describe a language translator?

A language translator such as a **compiler** or **interpreter** is needed to **convert** the **high level language** code into **machine code**. A computer understands only machine language.

💡 A translator turns **High** into **Machine** 🗣️➡️0101. Its two names: **C**ompiler and **I**nterpreter, "**C**an **I** translate?"

⚠️ **WATCH OUT**

A computer understands **only machine language**, never high level code directly. Spell it **Assembly** (not "assemble") and say "**level** language" for all three types.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which type of language uses only 0s and 1s?
2. Name the two kinds of language translator.
3. What is the process of creating a program called?

PART 3 · PROGRAMMING AND PYTHON

Lesson 8 · Meet Python: interactive and script mode

THE BIG IDEA

Python is a high level programming language with **simple syntax**, so it is easy to read and write. It was developed in **1991** in the **Netherlands**. You can use it in two ways: **interactive mode** (type a line, press Enter, see the answer at once) or **script mode** (write a long program in the editor and save it as a **.py** file). **IDLE** is the program that comes with Python.

KEY WORDS: **Python** · **syntax** · **Object Oriented** · **reusable code** · **interactive mode** · **script mode** · **.py** · **IDLE**

🐍 **IDLE Shell (interactive mode)**

```
>>> print("Hello")
Hello
>>> 5 + 3
8
```

Type, press **Enter**, get the output straight away

📄 **game.py (script mode)**

```
# my program
name = "Sara"
print("Hi", name)
print("Let's play!")
```

Write many lines, **save** as .py, then **Run**

	💬 Interactive mode	🎬 Script mode
Where	the Python shell (the >>> prompt)	the Python editor (IDLE)
Output	straight after you press Enter	when you run the program
Best for	very short lines of code	long programs with multiple lines
Save?	no	yes, as a **.py** file you can edit

Written questions (learn these answers)

Q14 **What do you know about Python?**

> Python is a **high level programming language**. It has **simple syntax** which makes it **easy to read and write**. It was developed in **1991** at the **National Research Institute for Mathematics and Computer Science** in the **Netherlands**.

💡 Python 🐍 hatched in **1991** (one year after HTML's 1990) in the **Netherlands** 🇳🇱 (think tulips 🌷). **Simple syntax** = easy to read and write.

Q15 **Write any 2 features of Python?**

> Python is **easy to learn and use** as compared to other programming languages.
> Python can perform **complex tasks** using **less code**.

💡 **Easy in, little out**: easy to learn and use; **less code** for complex tasks. ✌️

Q16 **What is an Object Oriented procedure?**

> Python supports Object Oriented procedure. An Object Oriented procedure helps the programmer to write **reusable code** and **develop applications** using **less code** as compared to other programming languages.

💡 Object Oriented = LEGO bricks 🧱: build a brick once and **reuse** it everywhere, so you build apps with **less code**.

Q17 **Describe Interactive Mode of Python while writing a code.**

> In the Interactive Mode when we type a command and **press Enter**, in the **very next step** we get the **output**. Interactive Mode is very convenient for writing **very short lines of code**.

💡 Interactive = chatting 💬: you type, press **Enter**, and Python answers **straight away**. Chats are short, so it suits **short lines**.

Q18 **Describe the script mode of Python language.**

> To write **long** Python codes or programs having **multiple lines**, script mode is the appropriate way. In this mode the code is written in the **Python editor** and saved with a **".py"** extension. In this mode we can **save and edit** the file.

💡 A **script** is a film script 🎬: **long**, many lines, written down and **saved** (as **.py**) so you can edit it later.

Q19 **What do you know about Python IDLE?**

> The **installer for Windows** installs an IDLE module **by default**. IDLE is an acronym for **Integrated Development and Learning Environment**. It is used in **script mode** for **creating, editing and executing** Python programs.

💡 IDLE is never idle! It **I**ntegrates, **D**evelops, **L**earns in an **E**nvironment, and it **C**reates, **E**dits and **E**xecutes scripts (C, E, E).

⚠️ **WATCH OUT**

Python is **1991**, HTML is **1990**: don't swap them. Python files end in **.py** (not .python), CSS files in **.css**. IDLE is **I**ntegrated **D**evelopment and **L**earning **E**nvironment.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. You want to write a 50-line game and keep it. Which mode should you use?
2. What does IDLE stand for?
3. Give one reason Python is easy to read and write.

PART 4 · DRAWING IN GIMP

Lesson 9 · GIMP painting tools

THE BIG IDEA

GIMP is a free program for drawing, painting and changing pictures. It has a large variety of **painting tools**, each with its own job: fill an area, mix two colours, paint soft strokes, draw sharp lines, or rub colour out.

KEY WORDS: GIMP · painting tools · bucket fill · gradient · paintbrush · pencil · airbrush · ink · eraser

	Painting tool	Its job
🪣	**Bucket fill**	fills colour in an area
	Gradient	fills a **mix of two colours** in an area (it does not darken)
🖌️	**Paintbrush**	paints with **fuzzy** (soft-edged) brush strokes; good for freehand drawing
✏️	**Pencil**	draws freehand lines with **hard edges**
💨	**Airbrush**	sprays soft colour, like a spray can
🖋️	**Ink**	draws like a **calligraphic ink pen**
🧽	**Eraser**	**removes areas of colour** from an image or a selected part



Paintbrush: fuzzy edge · Pencil: hard edge · Gradient: two colours mixed

From your worksheet

#	Tick the correct option
A10	**Which of the following is not a Painting Tool?** (a) Bucket fill (b) Airbrush ✔ **(c) Shear**
A12	**Which of the following is similar to a calligraphic ink pen?** (a) Smudge (b) Airbrush ✔ **(c) Ink tool**

#	Fill in the blanks
B11	The **Gradient** tool fills a mix of two colours in an area.
B12	The **Eraser** tool is used to remove areas of colour from an image.

#	Write T for true and F for false	Ans
C10	The Paintbrush tool is used to do freehand drawings.	T
C11	The Gradient tool is used to darken the image. Gradient mixes two colours. The tool that darkens is **Burn** (Lesson 10).	F

Written questions (learn these answers)

Q20 Why do we use GIMP for drawing?

> There are a **large variety of painting tools** in GIMP which can be used to **paint our drawings beautifully**. These tools can also be used to **draw, paint and modify images**.

💡 GIMP is a giant art box 🎨: **lots of painting tools** to **draw, paint and modify**. D-P-M: "**D**raw **P**retty **M**asterpieces".

Q21 **Enlist the names of some common tools used for drawings in GIMP?**

1. The **bucket fill** tool (fills colour in an area)
2. **Gradient** tool
3. **Paintbrush** tool
4. **Pencil** tool
5. **Eraser** tool

💡 **B**ig **G**reen **P**encils **P**aint **E**asily: **B**ucket fill 🪣, **G**radient 🌈, **P**aintbrush 🖌️, **P**encil ✏️, **E**raser 🧽.

Q22 **Why do we use the paintbrush tool in GIMP?**

The paintbrush tool allows us to **paint** using **fuzzy brush strokes**. All strokes are rendered using the **currently selected brush** type.

💡 A paintbrush is soft and **fuzzy** 🖌️ like a fluffy tail. Every stroke uses the brush **you selected**.

Q23 **What is the use of the pencil tool in drawing?**

The pencil tool is used to draw **free-hand lines** having **hard edges**. It is similar to the **paintbrush tool** but the pencil tool **does not produce fuzzy edges**.

💡 A pencil has a **sharp** point ✏️, so its lines have **hard edges**. The brush is the fuzzy one; the pencil is **not**.

Q24 **Is the Eraser tool important while drawing in GIMP?**

The eraser tool is a **very important** tool for drawing because it is used to **remove areas of colour** from an image or the **selected part** of an image.

💡 Everyone makes mistakes, so yes, it's **very important** 🧽: it **removes areas of colour** from the image or the selected part.

⚠️ **WATCH OUT**

Shear is a **transform** tool (it slants a picture), not a painting tool. **Gradient** mixes colours; it does **not** darken. Paintbrush = **fuzzy**, pencil = **hard** edges.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which tool is similar to a calligraphic ink pen?
2. Which tool fills a mix of two colours in an area?
3. You want a line with hard edges. Pencil or paintbrush?



PART 4 · DRAWING IN GIMP

Lesson 10 · GIMP fixing and effects tools

THE BIG IDEA

GIMP also has tools to **fix** and **change** photos. **Clone** copies one part of a picture onto another part; **Heal** removes defects like spots and scratches; **Dodge** lightens and **Burn** darkens; **Smudge** smears colour. **Fuzzy Select** selects areas of a similar colour, and **Filters** add special effects.

KEY WORDS **clone** **heal** **dodge** **burn** **smudge** **fuzzy select** **filters** **special effects**

	Tool	Its job
🐑	**Clone**	**copies** the selected part of an image and pastes it in another part
🩹	**Heal**	removes **defects** in images (spots, scratches)
☀️	**Dodge**	**lightens** part of the image
🔥	**Burn**	**darkens** part of the image

👆	**Smudge**	smears colours together, like a finger in wet paint
🪄	**Fuzzy Select**	**selects** touching (contiguous) areas of a similar colour, like a magic wand
✨	**Filters**	apply **special effects** to an image (blur, distort, artistic…)



Dodge ☀️ lightens | original | Burn 🔥 darkens

From your worksheet

#	Tick the correct option
A11	**Which of the following is used to copy a part of an image?** ✔ **(a) Clone** (b) Dodge (c) Both (a) and (b)
A13	**Which of the following is used to apply special effects to an image?** ✔ **(a) Filters** (b) Heal (c) Burn

#	Fill in the blanks
B10	The **Fuzzy Select** tool allows the selection of contiguous areas of a similar colour range.
B13	The **Heal** tool is used to remove defects in images.

#	Write T for true and F for false	Ans
C12	The Dodge tool is used to lighten the image.	T
C13	Filters are used to select objects in GIMP. Filters add special effects. Selecting is done by selection tools like **Fuzzy Select**.	F

Written questions (learn these answers)

Q25 **What is the use of the clone tool in GIMP?**

> In GIMP the clone tool **reproduces** the **selected parts** in an image. It **copies** the selected parts of the image and **pastes** them in **any other part** of the image.

💡 A **clone** is an identical twin 🐑🐑: the clone tool **copies** the selected part and **pastes** it somewhere else in the same image.

💡 **MEMORY TRICK**

Dodge = **D**aylight ☀️ (lighter); **B**urn = **B**lack toast 🍞 (darker). **Heal** heals a scratch like a plaster 🩹. **F**ilters = **F**un effects ✨.

⚠️ **WATCH OUT**

Only **Clone** copies (A11 is not "Both": Dodge just lightens). **Filters do not select** (C13 is False). Contiguous means "touching": Fuzzy Select picks touching areas of a similar colour.

✏️ **TRY THESE (ANSWERS AT THE BACK)**

1. Which tool removes defects such as spots and scratches?
2. Which tool darkens part of an image?
3. Which tool selects contiguous areas of a similar colour?

READ THIS THE NIGHT BEFORE

⚡ Cheat sheet

🌐 HTML

Hypertext Markup Language · **Tim Berners-Lee, 1990** · **W3C** sets web standards · text file, not case sensitive, many tags · HTML editors: Dreamweaver, UltraEdit, TextMate · text editors: Notepad, Notepad++ · view: double-click, or Edge + **Ctrl + O**

🏷️ Tags

inside **angular brackets** < > · opening `<HTML>`, closing `</HTML>` · both = **container tags** · order: HTML › HEAD › /HEAD › BODY › /BODY › /HTML

🖼️ Pictures

`<IMG>` adds a picture · attributes: src, **alt** (alternate text), height, width · formats: .jpg .gif **.png** .bmp · **.xlsx is not** a picture · **target is not** an IMG attribute

🔗 Links

types: **internal** and **external** (both) · **target** = where a link opens · **absolute URL** = complete path · colours: link, **alink** (active), **vlink** (visited) · C7: write **T**

🎨 CSS

Cascading Style Sheets · styles pages, does **not create** them · easy to format large websites · rule = **selector**, **property**, **value** · saved as **.css** · selectors pick HTML elements

📑 Style sheets

inline (style attribute, one element) · **internal** (in the head section) · **external** (separate .css file) · NOT "inside" · **font-family** = font · **text-indent** = first line · **background-position** = image position

🤖 Programs

program = sequence of instructions for a task · programming = creating programs · types: **machine**, **assembly**, **high** level · translator: **compiler** or **interpreter** → machine code · computer understands only machine language

🐍 Python

high level, simple syntax · **1991**, Netherlands · easy to learn, less code · Object Oriented → reusable code · **interactive**: Enter → output now, short code · **script**: long code, editor, **.py**, save and edit · **IDLE** = Integrated Development and Learning Environment

🖌️ Painting tools

🪣 bucket fill · 🌈 **gradient** = mix of two colours (not darken) · 🖌️ paintbrush = fuzzy, freehand · ✏️ pencil = hard edges · 💨 airbrush · 🖋️ **ink** = calligraphic pen · 🧽 **eraser** removes colour · **Shear is not** a painting tool

🪄 Fixing tools

🐑 **clone** copies a part · 🩹 **heal** removes defects · ☀️ **dodge lightens** · 🔥 burn darkens · 🪄 **fuzzy select** = similar colour areas · ✨ **filters** = special effects (they don't select)

🔑 Keyword → answer: the 25 written questions

Q1	**HTML**	Hypertext Markup Language · 1990 · Tim Berners-Lee
Q2	**use + HTML**	create web pages on the World Wide Web · hyperlinks
Q3	**features + HTML 5**	simple text files · not case sensitive · many tags
Q4	**HTML Editors + Text Editors**	HTML editors: Dreamweaver, UltraEdit, TextMate · text editors: Notepad, Notepad++
Q5	**view**	double-click · or Edge + Ctrl + O
Q6	**HTML tags**	elements, structure · < less than, > greater than = angular brackets
Q7	**opening + closing tags**	<HTML> opening, </HTML> closing · both = container tags
Q8	**CSS**	Cascading Style Sheets · styles HTML pages
Q9	**types of CSS**	inline · internal · external
Q10	**inline**	single HTML element · style attribute
Q11	**computer programming**	sequence of instructions · particular task · programming languages
Q12	**types + programming languages**	machine · assembly · high level
Q13	**language translator**	compiler or interpreter · high level → machine code
Q14	**Python**	high level · simple syntax · 1991 · Netherlands
Q15	**features of Python**	easy to learn and use · complex tasks with less code
Q16	**Object Oriented**	reusable code · applications with less code
Q17	**Interactive Mode**	press Enter → output at once · short lines
Q18	**script mode**	long code · Python editor · .py · save and edit
Q19	**Python IDLE**	Integrated Development and Learning Environment · script mode
Q20	**GIMP + drawing**	large variety of painting tools · draw, paint, modify
Q21	**common tools + drawings**	bucket fill · gradient · paintbrush · pencil · eraser
Q22	**paintbrush tool**	paint with fuzzy brush strokes · selected brush
Q23	**pencil tool**	free-hand lines, hard edges · not fuzzy
Q24	**Eraser tool**	very important · removes areas of colour
Q25	**clone tool**	copies selected parts · pastes them elsewhere

LEARN, COVER, WRITE, CHECK

All 25 model answers

The keyword tells you which answer to write. The **key phrases** score the marks. Cover the answer, write it on paper, then check.

Q1 What do you know about HTML?

HTML stands for **Hypertext Markup Language**. It was developed in the year **1990** by the scientist **Tim Berners-Lee**.

Q2 What's the use of HTML?

HTML is mainly used to **create web pages** that are displayed on the **World Wide Web**. Each page contains a series of **connections to other pages** called **hyperlinks**.

Q3 Write any three features of HTML 5?

1. HTML documents are **simple text files**.
2. HTML is **not a case sensitive** language.
3. HTML has **many tags** to make web pages.

Q4 Define HTML Editors and Text Editors.

We can create an HTML document using an **HTML editor** or any **text editor**.

HTML editors are **Adobe Dreamweaver**, **UltraEdit**, **TextMate**.

Text editors are **Notepad**, **Notepad++**.

We can create our HTML documents using Notepad.

Q5 How can we view the HTML document?

After creating and saving an HTML document we can view it by **double clicking** on the HTML document.

We can also open an HTML file by opening **Microsoft Edge** by clicking on its icon on the desktop or the taskbar and then pressing **Control + O** keys.

Q6 What do you know about HTML tags?

Tags are **elements** that are used to specify the **structure** of an HTML document. HTML tags begin with the **less than** character (<) and end with the **greater than** symbol (>), which are called **angular brackets**.

Q7 What is the difference between opening and closing tags?

In HTML, most tags have an **opening** and a **closing tag**. For example **<HTML>** is the opening tag and **</HTML>** is the closing tag. The tags which have both opening and ending tags are called **container tags**.

Q8 What do you know about CSS?

CSS stands for **Cascading Style Sheets**. CSS is used to **style HTML pages** and it's used to specify **how HTML elements are displayed**.

Q9 List down the different types of CSS?

1. **Inline** style sheet
2. **Internal** style sheet
3. **External** style sheet

Q10 Define inline style sheet.

The inline style sheet is used to apply the style to a **single HTML element** using the **style attribute**.

Q11 What do you know about computer programming?

A **sequence of instructions** given to the computer to perform a **particular task** is called a **program**. These programs are created using **programming languages**. The process of creating a program is also known as **programming**.

Q12 Write the names of types of computer programming languages?

There are three main types:

1. **Machine** level language
2. **Assembly** level language
3. **High** level language

Q13 Describe a language translator?

A language translator such as a **compiler** or **interpreter** is needed to **convert** the **high level language** code into **machine code**. A computer understands only machine language.

Q14 What do you know about Python?

Python is a **high level programming language**. It has **simple syntax** which makes it **easy to read and write**. It was developed in **1991** at the **National Research Institute for Mathematics and Computer Science** in the **Netherlands**.

Q15 Write any 2 features of Python?

Python is **easy to learn and use** as compared to other programming languages.

Python can perform **complex tasks** using **less code**.

Q16 What is an Object Oriented procedure?

Python supports Object Oriented procedure. An Object Oriented procedure helps the programmer to write **reusable code** and **develop applications** using **less code** as compared to other programming languages.

Q17 Describe Interactive Mode of Python while writing a code.

In the Interactive Mode when we type a command and **press Enter**, in the **very next step** we get the **output**. Interactive Mode is very convenient for writing **very short lines of code**.

Q18 Describe the script mode of Python language.

To write **long** Python codes or programs having **multiple lines**, script mode is the appropriate way. In this mode the code is written in the **Python editor** and saved with a **".py"** extension. In this mode we can **save and edit** the file.

Q19 **What do you know about Python IDLE?**

> The **installer for Windows** installs an IDLE module **by default**. IDLE is an acronym for **Integrated Development and Learning Environment**. It is used in **script mode** for **creating, editing and executing** Python programs.

Q20 **Why do we use GIMP for drawing?**

> There are a **large variety of painting tools** in GIMP which can be used to **paint our drawings beautifully**. These tools can also be used to **draw, paint and modify images**.

Q21 **Enlist the names of some common tools used for drawings in GIMP?**

> 1. The **bucket fill** tool (fills colour in an area)
> 2. **Gradient** tool
> 3. **Paintbrush** tool
> 4. **Pencil** tool
> 5. **Eraser** tool

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✏️ Answer key: Try these

Lesson 1 · What is HTML?

1. Tim Berners-Lee, in 1990.
2. Notepad and Notepad++ (WordPad also works).
3. **Ctrl + O** (Control and O).

Lesson 2 · Tags and page structure

1. `</HEAD>`
2. `</HTML>`
3. Container tags.

Lesson 3 · Pictures

1. The **alt** attribute.
2. .gif is a picture format; .xlsx is not (it is an Excel spreadsheet).
3. `<IMG src="sun.jpg">` (you can add alt, height and width too).

Lesson 4 · Links

1. **vlink**.
2. Relative (it is not the complete path).
3. Internal linking and external linking.

Lesson 5 · What is CSS?

1. Selector **p**, property **color**, value **red**.
2. **.css**
3. False: CSS styles web pages; HTML creates them.

Lesson 6 · Style sheets

1. Inline.
2. **text-indent**.
3. In the head section of the web page.

Lesson 7 · Programs and languages

1. Machine level language.
2. Compiler and interpreter.
3. Programming.

Lesson 8 · Python

1. Script mode (saved as a .py file).
2. Integrated Development and Learning Environment.
3. It has simple syntax.

Lesson 9 · Painting tools

1. The ink tool.
2. The gradient tool.
3. Pencil.

Lesson 10 · Fixing tools

1. The heal tool.
2. The burn tool.
3. The fuzzy select tool.